

FIBRE OPTICS PROBLEMS

Q1. Calculate the numerical aperture and hence the acceptance angle for an optical fiber. Given that the refractive indices of the core and the cladding are 1.45 and 1.40 respectively.

Given:- $\mu_1 = 1.45$; $\mu_2 = 1.40$

Formula :- $N.A. = \sin \theta_0 = \sqrt{\mu_1^2 - \mu_2^2}$

Solution:- $N.A. = \sqrt{1.45^2 - 1.40^2}$
 $= 0.3775$

$$\theta_0 = \sin^{-1}(N.A.) = \sin^{-1}(0.3775)$$
$$= 8.192^\circ$$

Ans:- N.A. of fiber is 0.3775 and its acceptance angle is 8.192°

Q2. The refractive index of core and cladding of a SI fiber are 1.52 and 1.41 respectively. Calculate (i) critical angle (ii) NA and (iii) maximum incidence angle.

Given:- $\mu_1 = 1.52$; $\mu_2 = 1.41$

Formula:- $\varphi_c = \sin^{-1} \frac{\mu_2}{\mu_1}$; $N.A. = \sin \theta_0 = \sqrt{\mu_1^2 - \mu_2^2}$

Solution:- $\varphi_c = \sin^{-1}\left(\frac{1.41}{1.52}\right) = 68.06^\circ$

$$N.A. = \sqrt{1.52^2 - 1.41^2} = 0.5677$$

$$\theta_0 = \sin^{-1}(0.5677)$$
$$= 34.59^\circ$$

Ans:- The critical angle is 68.06° and N.A. is 0.5677 and θ_0 is 34.59°

Q3. An optical fiber has a NA of 0.20 and refractive index of cladding is 1.59. Determine the core refractive index and the acceptance angle for the fiber in water which has a refractive index of 1.33.

Given:- NA=0.20; $\mu_2=1.59$; $\mu_o=1.33$

Formula:- $N.A. = \sin \theta_0 = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_o}$

Solution:- $N.A. = \sqrt{\mu_1^2 - \mu_2^2}$

$$\mu_1 = \sqrt{N.A.^2 + \mu_2^2} = \sqrt{0.2^2 + 1.59^2} = 1.6025$$

$$\theta_0 = \sin^{-1} \frac{N.A.}{\mu_o} = 8.64^\circ$$

Ans:- The R.I. of core is 1.6025 and acceptance angle 8.64°

Q4. A typical relative refractive index difference for an optical fiber is 1%. Estimated the numerical aperture and the critical angle at the core cladding interface if the core refractive index is 1.46.

Given:- $\Delta=0.01$; $\mu_1=1.46$

Formula:- $\Delta = \frac{\mu_1 - \mu_2}{\mu_1}$; $N.A. = \mu_1 \sqrt{2\Delta}$; $\varphi_c = \sin^{-1} \frac{\mu_2}{\mu_1}$

Solution:- $N.A. = 1.46(2 \times 0.01)^{1/2} = 0.2064$

$$\Delta = 1 - \frac{\mu_2}{\mu_1}$$

$$\frac{\mu_2}{\mu_1} = 1 - \Delta = 0.99$$

$$\varphi_c = \sin^{-1} 0.99 = 81.89^\circ$$

Ans:- The N.A. is 0.2064 and the critical angle is 81.89°.

Q5. A glass clad fiber is made with core glass of refractive index 1.5 and the cladding is doped to get a refractive index difference of

0.0005. Find [a] the refractive index of the cladding. [b] the critical internal reflection angle [c] Acceptance angle.

Given :- $\mu_1=1.5$; $\Delta= 0.0005$

Formula:- $\Delta= 1-\frac{\mu_2}{\mu_1}$; $N.A.=\sqrt{\mu_1^2 - \mu_2^2}$

Solution:- $\mu_2= \mu_1(1-\Delta)=1.5(1-0.0005) =1.49925$

$$\varphi_c = \sin^{-1} \frac{\mu_2}{\mu_1} = \sin^{-1} \frac{1.49925}{1.5} = 88.18^\circ$$

$$\begin{aligned} \theta_0 &= \sin^{-1} \sqrt{\mu_1^2 - \mu_2^2} \\ &= \sin^{-1} \sqrt{1.5^2 - 1.49925^2} = 2.718^\circ \end{aligned}$$

Ans:- The R.I of cladding is 1.49925 and the critical angle is 88.18° and the acceptance angle is 2.718°.

Q6. A step index fiber has core diameter $29 \times 10^{-6} \text{m}$. The refractive indices of the core and the cladding are 1.52 and 1.5189 resp. If the light of wavelength $1.3 \mu\text{m}$ is transmitted through the fiber, determine [a] Normalized frequency of the fiber. [b] The number of modes fiber will support.

Given :- $d=29 \times 10^{-6} \text{m}$; $\lambda = 1.3 \times 10^{-6} \text{m}$; $\mu_1= 1.52$; $\mu_2=1.5189$.

Formula :- $V = \frac{2\pi r \sqrt{\mu_1^2 - \mu_2^2}}{\lambda}$; $N_m = \frac{V^2}{2}$

Solution:- $V = \frac{3.14 \times 29 \times 10^{-6}}{1.3 \times 10^{-6}} \times \sqrt{(1.5)^2 - (1.5189)^2} = 4.049$

$$N_m = \frac{V^2}{2} = \frac{(4.049)^2}{2} = 8.197$$

Ans:- Normalized frequency is 4.049 and the number of modes is 8.